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Article

Single-Photon-Camera-Based Time and Spatially Resolved Electroluminescence Spectroscopy for Micro-LED Analysis

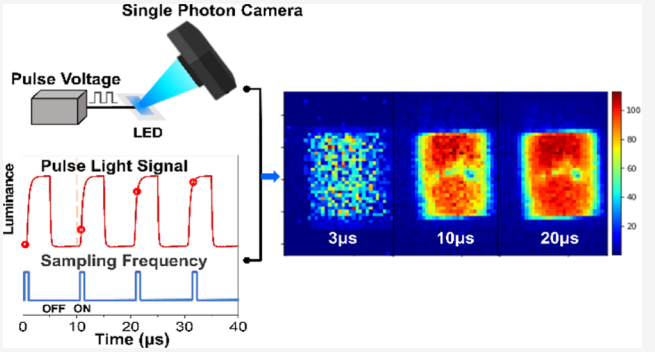
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Cite This: *ACS Photonics* 2024, 11, 731−736

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ABSTRACT: To investigate the operational mechanisms of micrometer-sized light-emitting diodes (micro-LEDs), we here demonstrate a transient methodology of time and spatially resolved electroluminescence spectroscopy (TSR-EL) to measure the spatial distribution of light emission from LED devices. By combining a single-photon camera (SPC) with the time-gated sampling method, we derived the time and spatially resolved electroluminescence intensity with increasing time. Benefiting from the high sensitivity of the SPC, this methodology can detect ultralow electroluminescence (EL) at the delay stage from the device operated around the turn-on voltage. Furthermore, we investigated the spatial light distribution of a typical quantum dots light-emitting diode (QLED) under different applied voltages and varied temperatures. It was found that the EL emission of the QLED device became more uniform with increasing temperature and applied voltage. Moreover, the methodology of TSR-EL is versatile to investigate other LEDs such as organic light-emitting diodes (OLEDs), micro-LEDs, etc.



KEYWORDS: *electroluminescence, spectroscopy, light-emitting diodes, time-gated sampling, quantum dot*

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INTRODUCTION

Motivated by the development of augmented reality display technology, it has been of great interest to develop micrometer-sized light-emitting diodes (micro-LEDs), including GaN-based LEDs, organic light-emitting diodes (OLEDs), and quantum dots light-emitting diodes (QLEDs).[1−3] Although great efforts have been made in materials and fabrication processing, there are still gaps between the device performance and industrial requirements.[4] For GaN-based blue micro-LEDs, the inductively coupled plasma etching process usually introduces side-wall defects, which induce a significant decrease in the external quantum efficiency (EQE) and nonuniformity.[5] For the micro-OLED display panel, the degradation of organic emitters strongly affects the device performance, which is of great importance to improve the operational lifetime.[6] Very recently, micro-QLED devices were also fabricated by applying the inkjet printing or photolithography process.[7,8] The decrease in device efficiency and nonuniformity of electroluminescence (EL) caused by size reduction and other effects are obstacles before industrialization.[9,10] To address these challenges, it has been much aspired to thoroughly characterize the performance of micro-LEDs in the time and spatial scales.

Time-resolved electroluminescence spectroscopy (TREL) is a nondestructive technique to investigate the operational mechanism of LED devices by recording the real-time brightness under a pulsed applied voltage.[11−13] A typical TREL setup usually contains a voltage-pulse generator, an oscilloscope, and a photodetector (avalanche photodiode, APD, or photomultiplier tube, PMT).[14] The voltage-pulse generator applies a square-wave voltage on the device to emit light, and then the emitted light is detected with a photodetector to record the EL process using an oscilloscope. Meanwhile, the current rise and decay can be recorded by adding a resistor to the device with an oscilloscope, which is termed as transient current (TRC—time-resolved current spectroscopy) measurement. By combining the TREL and TRC curves, a resistor−capacitor equivalent circuit model is proposed to illustrate the operating mechanism of LED devices by correlating the real-time brightness and current variation curves over time. More importantly, a few physical parameters can be derived to describe the operation of the LED device, including charge accumulation (Q), capacitance (C), exciton formation rate (r), and others.[15] These parameters are important to understand the degradation of the QLED device

Received: November 5, 2023
Revised: January 11, 2024
Accepted: January 11, 2024
Published: January 23, 2024



ACS Publications

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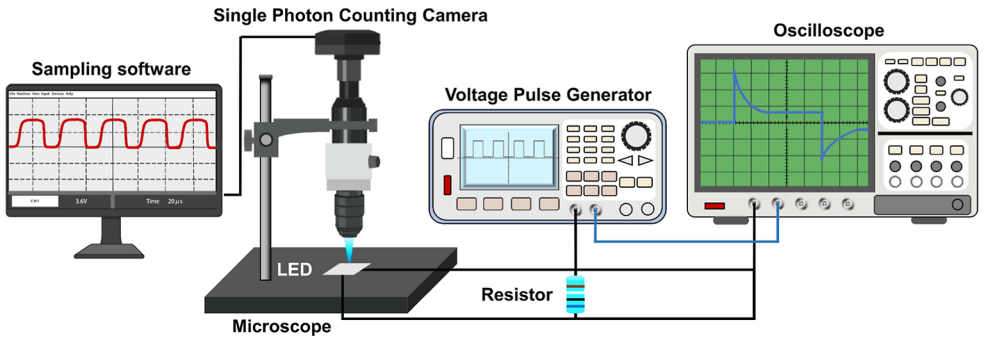


Figure 1. Schematic diagram of the TSR-EL system, including a voltage-pulse generator, an oscilloscope, a microscope, and an SPC.

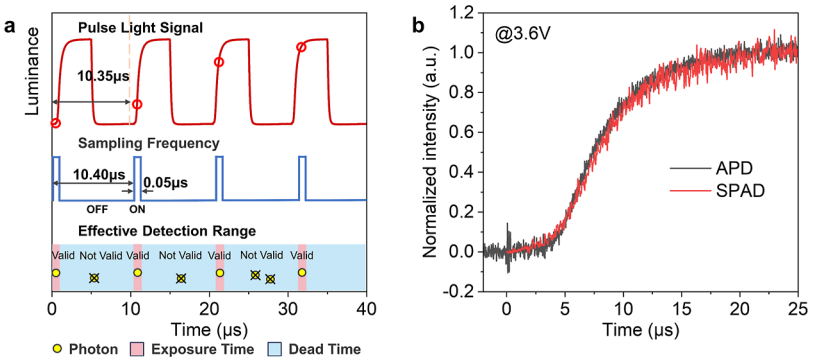


Figure 2. (a) Principle of the time-gated sampling method. (b) TREL curve (black line) of a typical OLED device using an APD detector and the TSR-EL curve (red line) derived from the same device using an SPAD detector under 3.6 V.

and predict its operational lifetime.[16] Limited by the detection ability of the APD and PMT, the conventional TREL setup is not able to recognize the weak signal at low brightness (below 300 cd/m^2). In addition, APD and PMT are single-pixel detectors without spatial resolution. By combining an InGaAs-based single-photon camera (SPC) with the time-gated sampling method, we here demonstrate a time and spatially resolved electroluminescence spectroscopy (TSR-EL) for characterizing micro-LEDs.

RESULTS AND DISCUSSION

Figure 1 schematically shows the TSR-EL setup, including a microscope with an SPC, a voltage-pulse generator, and an oscilloscope. Instead of the APD detector in the conventional TREL setup, an SPC (Micro Photon Devices S.R.L.-SPC[3]) is used to detect the spatial distribution of EL emission from the LED device. The SPC[3] is a two-dimensional imaging array of 64 × 32 smart pixels. Each pixel has a built-in single-photon avalanche diode (SPAD) detector. It can detect single photons by recording the avalanche current. When an avalanche current is detected by the active quenching circuit integrated on the same silicon chip, it quenches with the decrease of the voltage on the SPAD junction. Then, the SPAD resets to its initial state, ready to detect the next photon. This means that the frame sampling interval of an SPC is composed of two parts, including the exposure time and dead time, and the shortest frame sampling interval of commonly used SPCs is usually more than 10 μs. The SPC[3] camera has a minimum frame sampling interval of 10.4 μs during continuous snapshots. The sampling time interval is much larger than the step requirements of transient response of QLED or OLED (usually nanoseconds), which makes the detection of transient EL emission difficult using an SPC. If LED devices are stable within the time scale of transient testing, the TREL curves can be kept the same of multiple pulse periods. According to the previous reports on data sampling,[17,18] a time-gated sampling method is developed to derive the TREL curves of a stable device near the turn-on voltage using an SPC.

By setting the sampling interval of the camera's snapshot mode slightly longer than the pulse period time, the TREL curve can be reconstructed by reordering and extracting data from different periods. The specific testing procedure is as follows: we first set the sampling interval of the SPC to T, which includes the exposure time t_1 and the dead time t_2.

Here, $T = t_1 + t_2$, with $t_1 \ll t_2$, and the magnitude of t_1 is in the nanosecond range (to ensure sufficient sampling accuracy). To capture the full range of TREL curve, the period of the square-wave pulse voltage from the function generator is set to $n \times T -$ step, where n is a positive integer and step represents the sampling step size of the final TREL curve and is $\geq t_1$. Consequently, the first image represents the opening moment at $t = 0$ of the first period, the $(n + 1)$th image represents the t = step moment of the second period, the $(2n + 1)$th image represents the $t = 2 \times$ step moment of the third period, and so on. The $(kn + 1)$th image represents the $t = k \times$ step moment of the kth period. The SPC is configured to capture at least $[n \times (n \times T - \text{step})/\text{step}] + 1$ images. Finally, the data reconstruction is performed by selecting the images in a sequential order, starting from the $(1 + nk)$th image, where k takes values of 0, 1, 2, 3, and so on. Following the procedure, we achieved time-resolved EL curve measurements using an SPC. Furthermore, the time resolution of the obtained TREL curve can be adjusted by varying the camera's sampling interval and the period of the square-wave signal.

Figure 2a shows a typical example of the time-gated sampling method. The sampling time of our SPC is 10.4 μs,

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ACS Photonics 2024, 11, 731−736

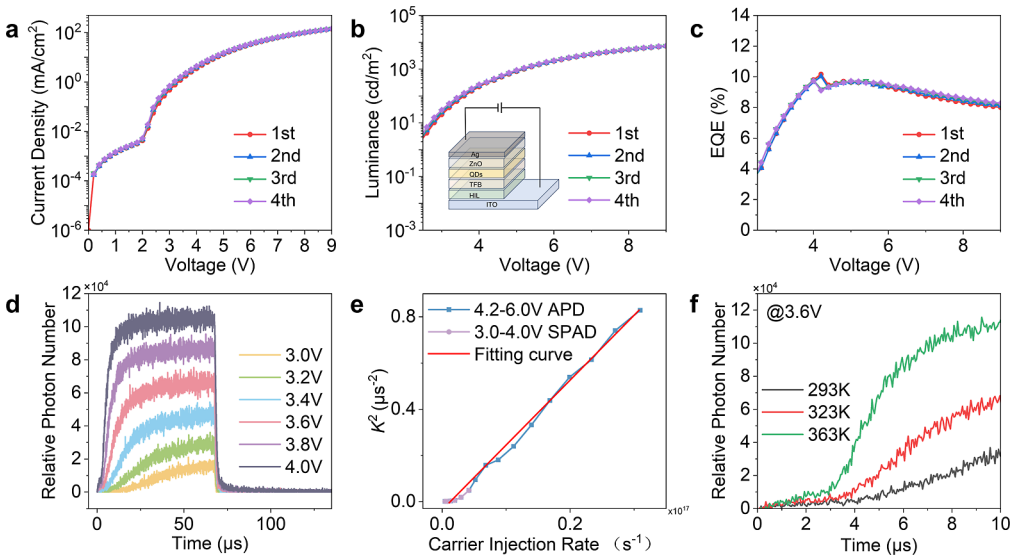


Figure 3. (a,b) Multiple current density−voltage−luminance (J−V−L) measurements of a typical QLED device. (c) EQE of the QLED device. (d) TREL curves obtained by the SPC from 3.0 to 4.0 V. (e) K^2−I relationship obtained by fitting the TREL curve of 3.0−4.0 V detected by SPAD (purple curve), K^2−I relationship obtained from 4.2 to 6.0 V detected by APD (blue curve), and the fitting curve of the relationship between K^2 and I (red curve). (f) Delay stage of TREL curves at different temperatures of 293, 323, and 363 K under 3.6 V.

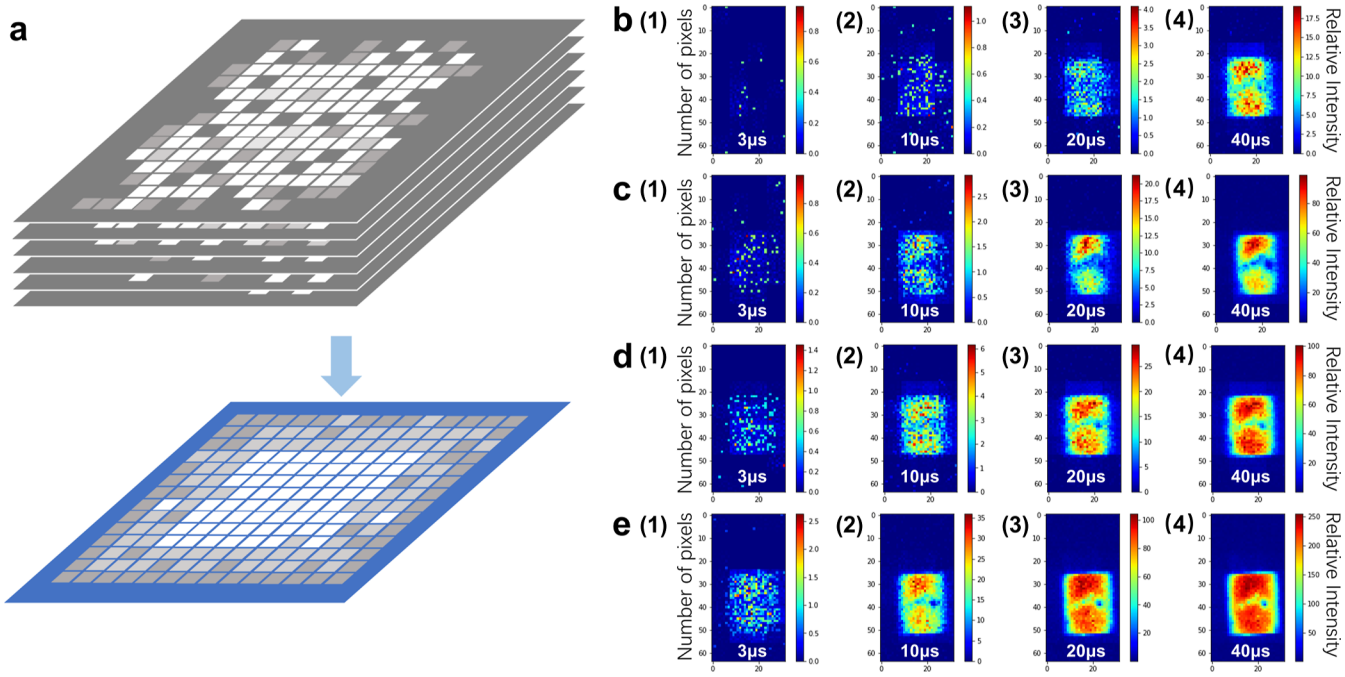


Figure 4. Time and spatially resolved pseudocolor images obtained by excitation of QLED under different pulse voltages and temperatures. (a) Schematic diagram of the pixel stacking method. QLED under excitation at different pulse voltages and temperatures: (b) 3.0 V, 293 K; (c) 3.0 V, 363 K; (d) 3.4 V, 293 K; (e) 3.4 V, 363 K.

the exposure time is set to 50 ns, and the dead time is 10.35 μs. We set the period of pulse voltage to 10.35 μs. Then, the information collected by the first picture contains the signal of the first 50 ns in the first period, the 50−100 ns of the second period is collected by the second photo, the 100−150 ns of the third period is collected by the third photo, and so on, until 207 photos are collected and restored to a full TREL period with a time resolution of 50 ns. To verify the effectiveness of the time-gated sampling method, we first performed the measurements of a highly stable OLED device. As shown in Figures 2b and S1 (see the Supporting Information), the detection effect of APD coincides almost exactly with the detection curve of SPAD, which confirms the reliability of this detection method.

We further applied the methodology to investigate QLED devices (the device structure is glass/ITO/HIL/HTL/QDs/ETL/Ag). Figure 3a−c shows the J−V−L and EQE curves of the tested QLED device. The device shows very good stability during the cyclic measurements. Figure 3d shows the TREL curve under low voltages from 3.0 to 4.0 V. By fitting the rising stage of the TREL curves with a prolonged time at applied voltages, we can derive the values of luminance increase rate

https://doi.org/10.1021/acsphotonics.3c01595
ACS Photonics 2024, 11, 731−736

(K) at different carrier injection rate (calculated from current). Figure 3e illustrates the relationship between the K^2 and the carrier injection rate, where the blue curve represents the fit results of the APD test and the purple curve represents the fit results of the SPAD test at lower voltages. Based on our previous theoretical derivation,[15] the relationship between K and I satisfies the following

$$K = \sqrt{I \times r}$$

The good linear relationship between K^2 and I supports the description of rising edge using the Langevin recombination model. It is also noticed that the use of SPAD in the TREL setup can expand the detection range of weak EL emission. Figure 3f shows the amplified curve of the delay stage. We noticed the weak EL emission at the delay stage, which is ignored in APD-based TREL measurement. To our knowledge, this phenomenon is reported for the first time. In the previous works, the delay stages of OLED and QLED were usually explained as the capacitor charging process[19,20] or slower carrier migration process from the electrode to the QD emission layer.[21] The observed EL intensity increase during the delay stage provides further information to analyze the physical mechanism in the delay stage. We studied the influence of carrier concentration on luminescence in the delay stage at different temperatures and voltages. As shown in Figure 3f, the slope of the delay curve increases with temperature under low voltage, meanwhile the delay time is shortened. While under high voltages, the delay curves are kept the same with increasing temperature (see Figure S2a). We further derived the total number of carriers participating in luminescence by integrating the area of the delay stage (see Figure S2b). It is found that the total number of carriers approach a threshold with prolonging of the time. Based on these results, we deduce that the device charging is coupled with pre-emitting. In addition, the pre-emitting is related to the carriers in the transport layer and/or the residual carriers at the interface within the delay stage.

We further illustrate the spatial measurement of the TSR-EL emission from an operational QLED. Figure 4a shows the pixel integration processing of the images acquired by the SPC. Figure 4b–e presents the time and spatially resolved emitting distributions of EL intensity for the same device under different voltages and temperatures. Under the conditions of increasing voltage, prolonged injection time, and higher temperature, the overall luminous intensity increases significantly.[22] To evaluate the uniformity of luminescence intensity in the images, we calculated the coefficient of variation by deriving the intensity value of each pixel in the images (see Figure S3).

As shown in Figure S3, the coefficient of variation generally decreases with increasing voltage (V), prolonged injection time (t), and higher temperature (Tem). These results suggest that the EL emission of a typical QLED device becomes more uniform due to the increase of the accumulation charge with time under high voltages and high temperatures. The TRC results (see Figure S4) also support that the charge accumulation increased with increasing applied voltage and temperature. In addition, we measured the TSR-EL curves and photos of a typical GaN-based blue micro-LED device (see Figure S5). A time resolution of 10 ns and a spatial resolution of 126 μm were achieved, showing the potential of TSR-EL to investigate the operational mechanism of micro-LEDs.

EXPERIMENTS

Device Information. The QLED devices are provided by TCL Corporate Research. The device has a structure of glass/ITO/HIL/HTL/QDs/ETL/Ag. OLED devices are provided by Prof. Ying Wang 's group at Technical Institute of Physics and Chemistry CAS. The device has a structure of Al(Liq)/ETL/HBL/EML/EBL/HTL/HIL/ITO. Micro-LED devices are provided by Prof. He Ding's group at Beijing Institute of Technology. The device has a structure of ITO/p+ GaN/EBL/MQWs/strained layers/n+ GaN/μ GaN/sapphire substrate.

Instruments. *TSR-EL and TRC.* The time-resolved detection system consists of a function generator (Siglent-SDG5162), an oscilloscope (Tektronix-DPO7104), an APD photodetector (Hamamatsu-C12702), an SPC (Micro Photon Devices S.R.L.-SPC^3), a microscope, a resistor, and several wires. The function generator is used to provide a period-adjustable pulse-voltage signal, the SPC and the APD photodetector are used to detect optical signals at low and high voltages, and the oscilloscope is used to identify and save data. The microscope is assembled by the laboratory itself. When detecting a device with a luminous area of 2 mm × 2 mm, the magnification is 2.4 times, and the working distance of the objective lens is 10.5 cm. When detecting a device with a luminous area of 50 μm × 80 μm, the magnification is 60 times, and the working distance of the objective lens is 7 mm.

J–L–V and EL Characteristics. The J–V–L test system consists of a source meter (Keithley-2400) and a spectrometer (Photo Research-PR788), and a fiber optic spectrometer (Beijing Avantes-AvaSpec HERO) is used to measure the voltage–current and current density–brightness characteristic curves of devices.

Test Procedure. *Step 1. Device Stability Verification.* Before the transient test, a voltage of 6 V is applied on the device for 5 min. Once the results of each J–L–V test remain the same, we start the TSR-EL measurement.

Step 2. Determining the Pulse Period. To gain full information on the TREL curves, the period of the square-wave pulse voltage is set to 135.15 μs (n × 10.4 − step, where n = 13 and step = 0.05 μs). The exposure time of the SPC is set to 50 ns, and 35,140 photos are taken at a time. The acquired photos are compressed and saved in TIFF format.

Step 3. Split Picture. A MATLAB program is derived to split the TIFF format image into 35,140 JPEG format photos; only the (13k + 1)th image is selected, a total of 2703 photos are obtained, and they are saved as raw data in a dedicated folder for subsequent processing (k = 0,1,2,...).

Step 4. TREL Curve Reconstruction. A Python program is derived to extract the total pixel intensity of the images in the folder mentioned in step 3 one by one, and initially a relationship curve between the intensity and the number of photo frames is drawn. Figure S6 shows two curve shapes. In all measurements, we first determined the number of frames corresponding to the voltage removal point, and then the starting point of the applied pulse voltage is found (frame ith). The data of one complete period after the starting point is what we need. For type I (see Figure S6a), the starting point of the pulse voltage is found half a period forward from the voltage removal point. Thus, one complete period data are frames from ith to (i + 2703)th. For type II (see Figure S6b), the starting point (frame ith) is found half a period after the voltage removal point, and we need to reorder the pictures by renaming them. The specific operation is renaming the photos

https://doi.org/10.1021/acsphotonics.3c01595
ACS Photonics 2024, 11, 731–736

after *i*th from 1 to 2703-*m*, while renaming the photos before *i*th from 2704-*m* to 2703. Thus, one complete period data are frames from 1st to 2703th, then drawing the relationship curve between the intensity and frames. Noise can be detected by the camera when the device is off. Figure S7 shows the results before denoising of Figures 3f and S2a. It can be seen that the relatively weak noise does not affect the conclusion.

Step 5. TSR-EL Processing Method. For type I, frames from *i*th to $(i + 2703)$th are collected for data analysis. For type II, frames from 1st to 2703th are collected for data analysis. The pictures are normalized with the arranged number and then integrated one by one. Finally, the output images are collected to show time and spatially resolved intensity distribution. The uniformity of device emission is assessed by calculating the coefficient of variation of the pixel intensity value matrix. The formula of the coefficient of variation is $C_v = \frac{\sigma}{\mu}$, where σ is the standard deviation of the pixel intensity value and μ is the mean value.

Step 6. Fitting Method of the TREL Rising Curve. In our previous work, we recognized that the carrier recombination process in the QLED device satisfies the Langevin recombination model and assumed that the charge injection is balanced and stable during the transient luminescence process, thus obtaining the fitting formula of the rising curve of TREL.

$$K = \sqrt{I \times r} \tag{1}$$

$$L = \tanh^2(K(t - T_d)) \tag{2}$$

The parameter r describes the rate at which excitons are formed per unit of free carriers in the emissive layer, and I = current/q represents the rate of carrier injection, obtained by conversion from the current value when the TRC curve is in a balance state. According to eq 2, we can fit the TREL curve to derive the parameter K, which represents the rate of brightness increment.

Methodology Applicability. The response of SPC[3] at a high flux and a low flux is not consistent. Figure S8 shows the relationship curve between the brightness results of the steady-state tester and the SPAD. The inflection point brightness is around 300 cd/m^2; beyond this value, the light detected by the SPAD tends to saturate, while below this value, the curve exhibits a generally linear trend. It shows that this technique is more suitable for detecting time-resolved EL under low brightness.

CONCLUSIONS

In this work, we have reported the TSR-EL technique by combining an SPC and a TREL setup, in which the dynamic luminescence process of QLEDs is described. A time-gated sampling approach is developed to achieve nanosecond resolution at lower EL intensity detection. (The minimum brightness value is approximately 20 cd/m^2.) This methodology enables us to detect the time and spatially resolved EL emission of a typical QLED operated at a turn-on voltage. The observed linear relationship between K^2 and the carrier injection rate indicates that a typical QLED can be described by the Langevin recombination model at low voltages, and we observe a slow emission phenomenon during the delay stage. We also investigated the spatially resolved TREL of QLEDs to illustrate the influence of applied voltage and temperature on the EL uniformity of QLED. Furthermore, we successfully detected the time and spatially resolved EL information on micro-LEDs, indicating the potential of the TSR-EL technique to investigate the mechanism of microdisplay devices in fast-switching scenarios. This is expected to further enhance their operational efficiency and stability, providing crucial technical support for the future development of the microdisplay.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsphotonics.3c01595.

OLED and micro-LED time and spatially resolved electroluminescence information (PDF)

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Author Contributions

H.Z. and L.B. conceived and supervised the project. Y.C., H.B., and L.P. carried out the experiments. Y.C. proposed the time-gated sampling method. Y.C., H.B., S.W., and H.Z. contributed to data analysis. Y.C., H.B., and L.P. wrote the code for data processing. Y.C., H.B., and H.Z. wrote the manuscript with subsequent input from other authors.

Funding

This work was supported by National Natural Science Foundation of China (62361166661, H.Z.) and the Science and Technology Development Fund, Macao SAR (file no. 0107/2023/AFJ, S.P.).

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We thank the MIIT Key Laboratory for Low-Dimensional Quantum Structure and Devices, Experimental Center of

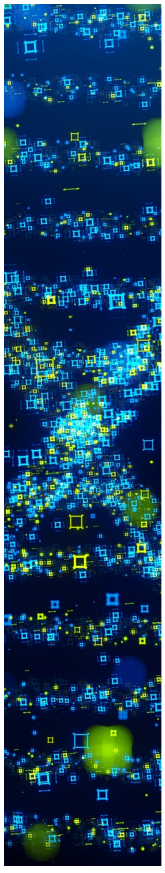
Advanced Materials of Beijing Institute of Technology, for technical assistance.

■ ABBREVIATIONS

QLED, quantum dots light-emitting diode; OLED, organic light-emitting diode; ICP, inductively coupled plasma; TREL, time-resolved electroluminescence spectroscopy; TRC, time-resolved current; TSR-EL, time and spatially resolved electroluminescence spectroscopy; SPAD, single-photon avalanche diode; APD, avalanche photodiode; PMT, photomultiplier tube; AQC, active quenching circuit

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https://doi.org/10.1021/acsphotonics.3c01595
ACS Photonics 2024, 11, 731−736